- Multivariate models

- General issues

- Practical example (ASREML)

So far: univariate models

- One dependent variable (trait; y)
- Fixed effects
- Random effects
  - Individual (direct) genetic effect
  - Permanent environment effect
  - Maternal (indirect) genetic effect
  - Common environment (maternal) effect
- Significance of effects; are they needed?
  - Wald test (F-statistic; fixed effects)
  - Log-Likelihood Ratio test (random effects)

Multivariate Models

- Joint analysis of two or more traits phenotypically and/or genetically correlated
  - Data on two or more phenotypes available on the same or genetically related individuals
- Optimal use of information
- Advantages:
  - Increases the accuracy of genetic predictions (BLUP)
  - Accounts for selection in the studied population:
    - Beef cattle culled due to poor weaning weight (ca. month 5)
    - Year 1 (12-mo) weight available for selected animals
    - Year 1 (yearling) weight analysis alone ➔ bias
    - Bivariate analysis weaning-yearling ➔ unbiasedness

Multivariate model

Univariate: y = Xb + Zu + e

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + \begin{bmatrix} Z_1 & 0 \\ 0 & Z_2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

y_i = vector of observations for the ith trait

b_i = vector of fixed effects for the ith trait

u_i = vector of random individual genetic effects for the ith trait

e_i = vector of random residual effects for the ith trait

Multivariate model

$$var\begin{bmatrix} u_1 \\ u_2 \\ e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} g_{11}A & g_{12}A & 0 & 0 \\ g_{21}A & g_{22}A & 0 & 0 \\ 0 & 0 & r_{11} & r_{12} \\ 0 & 0 & r_{21} & r_{22} \end{bmatrix}$$

$$X = \begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix}, Z = \begin{bmatrix} Z_1 & 0 \\ 0 & Z_2 \end{bmatrix}, \hat{b} = \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \end{bmatrix}, \hat{u} = \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \end{bmatrix} and\ y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Multivariate model: MME

$$
\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hline \hat{u}_1 \\ \hat{u}_2 \end{bmatrix}
=
\begin{bmatrix}
X'_1 r^{11} X_1 & X'_1 r^{12} X_2 & X'_1 r^{11} Z_1 & X'_1 r^{12} Z_2 \\
X'_2 r^{12} X_1 & X'_2 r^{22} X_2 & X'_2 r^{21} Z_1 & X'_2 r^{22} Z_2 \\
Z'_1 r^{11} X_1 & Z'_1 r^{12} X_2 & Z'_1 r^{11} Z_1 + A^{-1} g^{11} & Z'_1 r^{12} Z_2 + A^{-1} g^{12} \\
Z'_2 r^{21} X_1 & Z'_2 r^{22} X_2 & Z'_2 r^{21} Z_1 + A^{-1} g^{21} & Z'_2 r^{22} Z_2 + A^{-1} g^{22}
\end{bmatrix}^{-1}
\begin{bmatrix}
X'_1 r^{11} y_1 & X'_2 r^{12} y_2 \\
X'_2 r^{21} y_1 & X'_2 r^{22} y_2 \\
Z'_1 r^{11} y_1 & Z'_1 r^{12} y_2 \\
Z'_2 r^{21} y_1 & Z'_2 r^{22} y_2
\end{bmatrix}
$$

R=residual (co)variance matrix between traits $\begin{bmatrix} r_{11} r_{12} \\ r_{21} r_{22} \end{bmatrix}$

G=genetic (co)variance matrix between traits $\begin{bmatrix} g_{11} g_{12} \\ g_{21} g_{22} \end{bmatrix}$

Setting the MME

- Involves setting up X_1 for trait 1 and X_2 for trait 2

$$\mathbf{X}' R^{-1} \mathbf{X} = \begin{bmatrix} \mathbf{X}'_1 r^{11} \mathbf{X}_1 & \mathbf{X}'_1 r^{12} \mathbf{X}_2 \\ \mathbf{X}'_2 r^{21} \mathbf{X}_1 & \mathbf{X}'_2 r^{22} \mathbf{X}_2 \end{bmatrix}$$

- Note that $X_1'r^{11}X_1$, can be set up by forming initially $X_1'X_1$ and multiplying each element by r^{11}, i.e., the element of R^{-1} corresponding to trait one; same for the other traits.

Multivariate model: MME

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{u}_1 \\ \hat{u}_2 \end{bmatrix} = \begin{bmatrix} X'_1 r^{11} X_1 & X'_1 r^{12} X_2 & X'_1 r^{11} Z_1 & X'_1 r^{12} Z_2 \\ X'_2 r^{12} X_1 & X'_2 r^{22} X_2 & X'_2 r^{21} Z_1 & X'_2 r^{22} Z_2 \\ Z'_1 r^{11} X_1 & Z'_1 r^{12} X_2 & Z'_1 r^{11} Z_1 + A^{-1} g^{11} & Z'_1 r^{12} Z_2 + A^{-1} g^{12} \\ Z'_2 r^{21} X_1 & Z'_2 r^{22} X_2 & Z'_2 r^{21} Z_1 + A^{-1} g^{21} & Z'_2 r^{22} Z_2 + A^{-1} g^{22} \end{bmatrix}^{-1} \begin{bmatrix} X'_1 r^{11} y_1 & X'_2 r^{12} y_2 \\ X'_2 r^{21} y_1 & X'_2 r^{22} y_2 \\ Z'_1 r^{11} y_1 & Z'_1 r^{12} y_2 \\ Z'_2 r^{21} y_1 & Z'_2 r^{22} y_2 \end{bmatrix}$$

Univariate model: MME

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

Multivariate model: MME

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{u}_1 \\ \hat{u}_2 \end{bmatrix} = \begin{bmatrix} X'_1 r^{11} X_1 & X'_1 r^{12} X_2 & X'_1 r^{11} Z_1 & X'_1 r^{12} Z_2 \\ X'_2 r^{12} X_1 & X'_2 r^{22} X_2 & X'_2 r^{21} Z_1 & X'_2 r^{22} Z_2 \\ Z'_1 r^{11} X_1 & Z'_1 r^{12} X_2 & Z'_1 r^{11} Z_1 + A^{-1} g^{11} & Z'_1 r^{12} Z_2 + A^{-1} g^{12} \\ Z'_2 r^{21} X_1 & Z'_2 r^{22} X_2 & Z'_2 r^{21} Z_1 + A^{-1} g^{21} & Z'_2 r^{22} Z_2 + A^{-1} g^{22} \end{bmatrix}^{-1} \begin{bmatrix} X'_1 r^{11} y_1 & X'_2 r^{12} y_2 \\ X'_2 r^{21} y_1 & X'_2 r^{22} y_2 \\ Z'_1 r^{11} y_1 & Z'_1 r^{12} y_2 \\ Z'_2 r^{21} y_1 & Z'_2 r^{22} y_2 \end{bmatrix}$$

Univariate model: MME

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

$$\lambda = \frac{\sigma_e^2}{\sigma_u^2}$$

R=residual (co)variance matrix between traits $\begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}$

G=genetic (co)variance matrix between traits $\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$

Example - 2 traits

Individual	gender	sire	dam	WWG	PWG
4	male	1	-	4.5	6.8
5	female	3	2	2.9	5.0
6	female	1	2	3.9	6.8
7	male	4	5	3.5	6.0
8	male	3	6	5.0	7.5

$$\mathbf{G} = \begin{array}{c} \\ \text{WWG} \\ \text{PWG} \end{array} \begin{array}{c} \begin{array}{cc} g_{11} & g_{12} \end{array} \\ \begin{bmatrix} 20 & 18 \\ 18 & 40 \end{bmatrix} \end{array} \quad \text{and} \quad \mathbf{R} = \begin{array}{c} \\ \text{WWG} \\ \text{PWG} \end{array} \begin{array}{c} \begin{array}{cc} r_{11} & r_{12} \end{array} \\ \begin{bmatrix} \mathbf{40} & \mathbf{11} \\ \mathbf{11} & \mathbf{30} \end{bmatrix} \end{array}$$

Design Matrices

Inverses are:

$$\mathbf{G}^{-1} = \begin{bmatrix} \overset{g^{11}}{0.084} & \overset{g^{12}}{-0.038} \\ \overset{g^{21}}{-0.038} & \overset{g^{22}}{0.042} \end{bmatrix} \text{ and } \mathbf{R}^{-1} = \begin{bmatrix} \overset{r^{11}}{0.028} & \overset{r^{12}}{-0.010} \\ \overset{r^{21}}{-0.010} & \overset{r^{22}}{0.037} \end{bmatrix}$$

Design matrices:

$$\mathbf{X}_1' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \qquad \mathbf{X}_2' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

MME

$$\mathbf{X}_1'\mathbf{X}_1 r^{11} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} 0.028 = \begin{pmatrix} 0.084 & 0 \\ 0 & 0.056 \end{pmatrix}$$

$$\mathbf{X}_2'\mathbf{X}_2 r^{22} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} 0.037 = \begin{pmatrix} 0.111 & 0 \\ 0 & 0.074 \end{pmatrix}$$

$$\mathbf{X}_2'\mathbf{X}_1 r^{21} = \mathbf{X}_1'\mathbf{X}_2 r^{12} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} -0.01 = \begin{pmatrix} -0.03 & 0 \\ 0 & -0.02 \end{pmatrix}$$

$$\mathbf{X}'\mathbf{R}^{-1}\mathbf{X} = \begin{bmatrix} \mathbf{X}'_1 r^{11} \mathbf{X}_1 & \mathbf{X}'_1 r^{12} \mathbf{X}_2 \\ \mathbf{X}'_2 r^{21} \mathbf{X}_1 & \mathbf{X}'_2 r^{22} \mathbf{X}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{0.084} & \mathbf{0.000} & \mathbf{-0.03} & \mathbf{0.00} \\ \mathbf{0.00} & \mathbf{0.056} & \mathbf{0.00} & \mathbf{-0.02} \\ \mathbf{-0.03} & \mathbf{0.00} & \mathbf{0.111} & \mathbf{0.00} \\ \mathbf{0.00} & \mathbf{-0.02} & \mathbf{0.00} & \mathbf{0.074} \end{bmatrix}$$

Design matrices - random

$Z'_1 Z_1 r^{11}$ = diag (0. 0. 0. 0.028 0.028 0.028 0.028 0.028)

$Z'_2 Z_2 r^{22}$ = diag (0. 0. 0. 0.037 0.037 0.037 0.037 0.037)

$Z'_2 Z_1 r^{21}$ = $Z'_1 Z_2 r^{12}$ = diag (0. 0. 0. -0.01 -0.01 -0.01 -0.01 -0.01)

Design matrices - random

$Z'R^{-1}Z$ is

0.000										
0	0.000									
0	0	0.000								
0	0	0	0.028							
0	0	0	0.0	0.028						
0	0	0	0.0	0.0	0.028					
0	0	0	0.0	0.0	0.0	0.028				
0	0	0	0.0	0.0	0.0	0.0	0.028			
0	0	0	0.0	0.0	0.0	0.0	0.0	0.000		
0	0	0	0.0	0.0	0.0	0.0	0.0	0	0.000	
0	0	0	0.0	0.0	0.0	0.0	0.0	0	0	0.000
0	0	0	-0.01	0.0	0.0	0.0	0.0	0	0	0
0	0	0	0.0	-0.01	0.0	0.0	0.0	0	0	0
0	0	0	0.0	0.0	-0.01	0.0	0.0	0	0	0
0	0	0	0.0	0.0	0.0	-0.01	0.0	0	0	0
0	0	0	0.0	0.0	0.0	0.0	-0.01	0	0	0

0.037					
0	0.037				
0	0	0.037			
0	0	0	0.037		
0	0	0	0	0.037	
0	0	0	0	0	0.037

Effect solutions, (co)variance component estimation

Effects	Traits	
Sex	WWG	PWG
1	0.964	0.920
2	0.000	0.000

Individuals		
1	0.151	0.280
2	-0.015	-0.008
3	-0.078	-0.170
4	-0.010	-0.013
5	-0.270	-0.478
6	0.276	0.517
7	-0.316	-0.479
8	0.244	0.392

$$R = \begin{array}{c} WWG \\ PWG \end{array} \begin{bmatrix} 40 & 11 \\ 11 & 30 \end{bmatrix}$$

Residual correlation = 0.32

$$G = \begin{array}{c} WWG \\ PWG \end{array} \begin{bmatrix} 20 & 18 \\ 18 & 40 \end{bmatrix}$$

Genetic correlation = 0.64

Effect solutions

Multivariate analysis			Univariate analysis	
Effects	Traits		Traits	
Sex	WWG	PWG	WWG	PWG
1	0.964	0.920	0.954	0.919
2	0.000	0.000	0.000	0.000
Individuals				
1	0.151	0.280	0.098	0.277
2	-0.015	-0.008	-0.019	-0.005
3	-0.078	-0.170	-0.041	-0.171
4	-0.010	-0.013	-0.009	-0.013
5	-0.270	-0.478	-0.186	-0.471
6	0.276	0.517	0.177	0.514
7	-0.316	-0.479	-0.249	-0.464
8	0.244	0.392	0.183	0.384

Consistent results are generally expected

Not the same

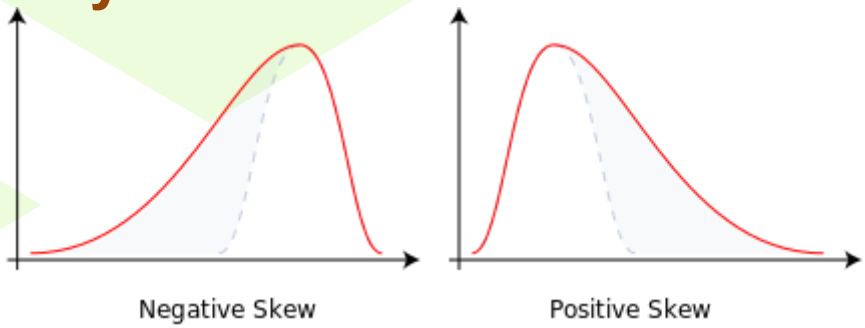
Multivariate analysis may:

- **Increase accuracy (B in BLUP)**
- **Underpin unbiasedness (U in BLUP)**

General Issues

General issues to consider in data analysis and model definition:

- **Descriptive statistics of the data to gain an understanding of the distribution of the data.**
  - **Such summary statistics may include, the mean, standard deviations, highest and minimum values to gain an understanding of outliers, skewness, kurtosis**
  - **Normality tests**



  - **If necessary carry out edits and/or data transformations**

General Issues

- Defining and testing fixed effects
  - some may be central to the analysis
  - others are “nuisance” factors which must be accounted for
  - interactions should be tested
  - Wald test (conditional F-statistic)

General Issues

- Defining and testing random effects
  - Individuals (humans, animals, plants, trees…..)
    - Genetic effect
  - Repeated measurements
    - Permanent environment
  - Common environment
  - Maternal effects
    - Genetic
    - Environmental
  - (log-)likelihood ratio test

General Issues

- Using information from above steps to put together the appropriate linear mixed model for the analysis data to
  - address your hypothesis
  - derive BLUEs and BLUPs
    - Genetic/gene effects (evolution, medicine, animal/plant breeding)
    - Genetic/genomic evaluations (anima/plant breeding)

Example – input file

Individ	father	mother	sex	ethnicity	bodyweight	bodyfat	LDL	Glucose
500001	1378	10001	Female	2	79	48	4.51	12.89
500002	1379	10002	Female	3	70.5	42.5	3.49	9.81
500003	1379	10002	Male	3	72	49	3.79	7.80
500004	1380	10003	Male	2	86	53.5	5.56	7.28
500005	1381	10004	Male	1	85	59.5	4.59	11.08
500006	1382	10005	Male	1	81.5	52.5	3.34	7.14
500007	1383	10006	Female	1	85.5	53.5	3.47	9.99
500009	1384	10007	Male	2	83	58.5	4.88	5.62
500010	1385	10008	Male	2	78.5	51	5.27	6.85
500011	1386	10009	Female	3	70.5	38.5	3.69	5.61

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ca. 4,000 records

LDL = Low-density lipoproteins ("bad" cholesterol)

Example script (*myscript.as*)

```
Univariate analysis - testing
 individual !P    # direct genetic effect - linked to pedigree
 father !P        # linked to pedigree
 mother !P        # linked to pedigree
 sex !A2          # fixed effect
 ethnicity 4      # fixed effect
 bodyweight       # fixed effect - covariate (could potentially be trait of study)
 bodyfat          # fixed effect - covariate (could potentially be trait of study)
 LDL              # trait of study
 Glucose          # trait of study

metabolic.txt !SKIP 1                                        #Pedigree file
metabolic.txt !SKIP 1 !MAXITER 100 !TYPEIII !FCON            #Data file

 LDL ~ mu  sex  ethnicity  bodyweight  bodyfat
```

Example output (*myscript.asr*)

```
*****************************************************
*****************************************************
*****************************************************
*****************************************************
*****************************************************
*****************************************************
* Contact support@asreml.co.uk for licensing and support  *
 *****************************************************
 Folder: /home/gbanos/
  individual !P
  father !P
  mother !P
  sex !A2
 Reading pedigree file metabolic.txt : skipping  1  lines
 Using an adapted version of  Meuwissen & Luo GSE 1992 305-313:  Specify !METHOD 1 for column method.
 PEDIGREE [metabolic.txt ] has     9186 identities,   19878 Non zero elements
 QUALIFIERS: !SKIP 1 !MAXITER 20 !TYPEIII
 Reading metabolic.txt  FREE FORMAT skipping     1 lines

 Univariate analysis of LDL
 Using     4066 records of    4066 read
```

Example output (*myscript.asr*)

```
Model term                  Size    #miss   #zero   MinNon0  Mean     MaxNon0
 1 individual   !P          9186    0       0       3.000    4390.    9186.
 2 father       !P          9186    0       0       2.000    4336.    9185.
 3 mother       !P          9186    0       0       1.000    4335.    9184.
 4 sex                      2       0       0       1        1.3251   2
 5 ethnicity                4       0       0       1        2.3492   4
 6 bodyweight               0       0       42.00   70.49    92.00
 7 bodyfat                  0       0       22.50   47.72    80.00
 8 LDL          Variate     0       0       1.043   4.018    7.720
 9 Glucose                  0       0       2.054   8.630    15.18
10 mu                       1
Forming      9 equations:   9 dense.
Initial updates will be shrunk by factor    0.224
Notice:      2 singularities detected in design matrix.
 1 LogL=-1821.56    S2= 0.89025     4059 df    1.000
Final parameter values                  1.0000
```

Example output (*myscript.asr*)

- - - Results from analysis of LDL - - -

Source	Model terms	Gamma	Component	Comp/SE	% C
Variance	4066 4059	1.00000	0.890253	45.05	0 P

Wald F statistics

Source of Variation	NumDF	DenDF_con	F-inc	F-con M	P-con
10 mu	1	4059.0	73720.02	66.26 .	<.001
4 sex	1	4059.0	1261.91	920.81 A	<.001
5 ethnicity	3	4059.0	5.31	2.98 A	0.031
6 bodyweight	1	4059.0	0.65	1.50 A	0.224
7 bodyfat	1	4059.0	78.35	78.35 A	<.001

Notice: The DenDF values are calculated ignoring fixed/boundary/singular variance parameters using algebraic derivatives.

	Solution	Standard Error	T-value	T-prev
7 bodyfat				
1	0.175171E-01	0.197903E-02	8.85	
6 bodyweight				
1	-0.348889E-02	0.285168E-02	-1.22	
5 ethnicity				
2	-0.760818E-01	0.538250E-01	-1.41	
3	-0.954510E-01	0.366526E-01	-2.60	-0.39
4	0.806109E-01	0.115566	0.70	1.57
4 sex				
Male	1.05137	0.346474E-01	30.34	
10 mu				
1	3.14914	0.208855	15.08	

Finished: ****[date time]**** LogL Converged

Example solutions (*myscript.sln*)

bodyfat	1	0.1752E-01	0.1979E-02
bodyweight	*1*	*-0.3489E-02*	*0.2852E-02*
ethnicity	1	0.000	0.000
ethnicity	2	-0.7608E-01	0.5383E-01
ethnicity	3	-0.9545E-01	0.3665E-01
ethnicity	4	0.8061E-01	0.1156
sex	Female	0.000	0.000
sex	Male	1.051	0.3465E-01
mu	1	3.149	0.2089

Example script (*myscript.as*)

```
Univariate analysis - testing
 individual !P     # direct genetic effect - linked to pedigree
 father !P         # linked to pedigree
 mother !P         # linked to pedigree
 sex !A2           # fixed effect
 ethnicity 4       # fixed effect
 bodyweight        # fixed effect - covariate OR trait of study
 bodyfat           # fixed effect - covariate OR trait of study
 LDL               # trait of study
 Glucose           # trait of study

metabolic.txt !SKIP 1                                  #Pedigree file
metabolic.txt !SKIP 1 !MAXITER 100 !TYPEIII !FCON      #Data file

 LDL ~ mu   sex   ethnicity   bodyweight   bodyfat   !r individual
```

Example output

```
 1 LogL=-1813.52    S2= 0.80708    4059 df   0.1000    1.000
 2 LogL=-1811.98    S2= 0.78320    4059 df   0.1336    1.000
 3 LogL=-1810.16    S2= 0.74301    4059 df   0.1959    1.000
 4 LogL=-1809.32    S2= 0.70737    4059 df   0.2581    1.000
 5 LogL=-1809.16    S2= 0.68588    4059 df   0.2993    1.000
 6 LogL=-1809.16    S2= 0.68364    4059 df   0.3037    1.000
 7 LogL=-1809.16    S2= 0.68349    4059 df   0.3040    1.000
Final parameter values                0.30406    1.0000
      - - - Results from analysis of LDL - - -
      Approximate stratum variance decomposition
Stratum     Degrees-Freedom   Variance     Component Coefficients
individual        3564.54   0.915404        1.1     1.0
Residual Variance   494.46   0.683475        0.0     1.0

Source            Model  terms    Gamma  Component   Comp/SE   % C
individual         9186   9186  0.304056   0.207818    4.77    0 P
Variance           4066   4059  1.00000    0.683475   15.72    0 P
```

Example genetic parameters

Source	Model	terms	Gamma	Component	Comp/SE	% C
individual	9186	9186	0.304056	0.207818	4.77	0 P
Variance	4066	4059	1.00000	0.683475	15.72	0 P

F *vu* 1 #3

F *ve* 2 #4

F *vp* 1+2 #5

H h^2 3 5

ASReml ********** Univariate analysis - testing

myscript.pvc created *****[date time]*****

3 vu	1	0.2078	0.4353E-01
4 ve	2	0.6835	0.4347E-01
5 vp	1	0.8913	0.1995E-01

h^2 = vu 1 3/vp 1 5= 0.2332 0.0479

Notice: The parameter estimates are followed by their approximate standard errors.

Assessment of random effect

- Same fixed effects
- $D = 2\log(L_{r2}/L_{r1}) = 2[\log(L_{r2})-\log(L_{r1})]$
  - Chi-square with d.f. = $r_2 - r_1$

Model	Log L	No. parameters
1 (Fixed effects only)	-1821.56	1
2 (Mixed)	-1809.16	2

- D = 24.8
- Chi-square df=1, 0.05 → 3.84

Example script (*myscript.as*)

```
Bivariate analysis - testing
 individual !P      # direct genetic effect - linked to pedigree
 father !P          # linked to pedigree
 mother !P          # linked to pedigree
 sex !A2            # fixed effect
 ethnicity 4        # fixed effect
 bodyweight         # fixed effect - covariate OR trait of study
 bodyfat            # fixed effect - covariate OR trait of study
 LDL                # trait of study
 Glucose            # trait of study

metabolic.txt !SKIP 1                                    #Pedigree file
metabolic.txt !SKIP 1 !MAXITER 100 !TYPEIII !FCON        #Data file

 LDL  Glucose ~ Trait  Tr.sex  Tr.ethnicity   Tr.bodyweight   Tr.bodyfat
                !r  Tr.individual
```

Example script (*myscript.as*)

```
LDL  Glucose ~ Trait  Tr.sex  Tr.ethnicity   Tr.bodyweight   Tr.bodyfat
                !r  Tr.individual

1 0 1
0 0 0

Trait 0 US !GP !+3
 1.0
 0.0  1.0

Tr.individual 2
2 0 US !+3
1.0
0.0  1.0
individual  0  AINV
```

Starting values:
Variance_1
Covariance Variance_2

Could be random
Hint: use Variance from univariate analyses

Example output (*myscript.asr*)

```
1 LogL=-8358.23    S2=  1.0000      6518 df   :  2 components constrained
2 LogL=-7626.28    S2=  1.0000      6518 df   :  2 components constrained
3 LogL=-6824.38    S2=  1.0000      6518 df   :  2 components constrained
4 LogL=-6439.95    S2=  1.0000      6518 df
5 LogL=-6203.05    S2=  1.0000      6518 df
6 LogL=-6126.65    S2=  1.0000      6518 df
7 LogL=-6122.24    S2=  1.0000      6518 df
8 LogL=-6122.21    S2=  1.0000      6518 df
9 LogL=-6122.21    S2=  1.0000      6518 df
```

Source	Model	terms	Gamma	Component	Comp/SE	% C
Residual	UnStructured	1 1	0.651262	0.651262	11.78	0 P
Residual	UnStructured	2 1	-0.822702E-01	-0.822702E-01	-0.77	0 P
Residual	UnStructured	2 2	5.55519	5.55519	14.02	0 P
Tr.individual	UnStructured	1 1	0.244539	0.244539	4.37	0 U
Tr.individual	UnStructured	2 1	0.358476	0.358476	3.45	0 U
Tr.individual	UnStructured	2 2	0.840025	0.840025	2.20	0 U

Example script (*myscript.as/.pin*)

```
F vu1    4       #7
F vu12   5       #8
F vu2    6       #9

F ve1    1       #10
F ve12   2       #11
F ve2    3       #12

F vp1 7+10       #13
F vp12 8+11      #14
F vp2 9+12       #15

H h²1    7   13
H h²2    9   15

R ru12    7    8    9
R re12   10   11   12
R rp12   13   14   15
```

Example genetic parameters

```
F vu1    4      #7        ASReml ****************      Bivariate analysis - testing
F vu12   5      #8            myscript.pvc created *****[date time]*****
F vu2    6      #9         7 vu1  4      0.2445      0.5595E-01
                           8 vu12  5     0.3585      0.1040
F ve1    1      #10        9 vu2  6      0.8400      0.3817
F ve12   2      #11       10 ve1  1      0.6513      0.5530E-01
F ve2    3      #12       11 ve12  2    -0.8227E-01  0.1066
                          12 ve2  3      5.555       0.3963
F vp1 7+10      #13       13 vp1  7      0.8958      0.2236E-01
F vp12 8+11     #14       14 vp12  8     0.2762      0.4696E-01
F vp2 9+12      #15       15 vp2  9      6.395       0.1587
                            h²1        = vu1  4    7/vp1  7    13=        0.2730    0.0613
H h²1   7   13              h²2        = vu2  6    9/vp2  9    15=        0.1314    0.0594
H h²2   9   15              ru12       = vu12  5 /SQR[vu1  4  *vu2  6  ]=   0.7909    0.2642
                            re12       = ve12  2 /SQR[ve1  1  *ve2  3  ]=  -0.0433    0.0566
R ru12    7    8    9       rp12       = vp12  8 /SQR[vp1  7  *vp2  9  ]=   0.1154    0.0193
R re12   10   11   12     Notice: The parameter estimates are followed by
R rp12   13   14   15          their approximate standard errors.
```

Genetic evaluations (BLUP)

- Genetic evaluation involves the estimation of EBVs and associated reliabilities or accuracies.
- This usually occurs at the national level
  - Productive, health, fertility et al traits in dairy/beef cattle, sheep and goats in the UK
    - National Unit at SRUC in Edinburgh
    - Most other countries
  - For poultry, pigs and fish genetic evaluations tend to be calculated within breeding programmes
- Different steps

Steps in genetic evaluations

- Input data validation/edits
  - This is to ensure observations (data, records) are within reasonable (biological) range
  - No large unexplained changes in data in the current evaluation compared to previous evaluation
  - Consistencies in pedigree (e.g. ancestors born before progeny etc.)
  - Do assumptions hold (normal distribution of data)?
- Use of appropriate software to estimate variance components and genetic parameters (e.g. ASREML, R, DMU etc.)
  - Pertain to trait
  - Computationally demanding
  - Possible additional data edits ➔ most informative subset of data (e.g. ancestors with more than a min. no. progeny)
  - Beware of selection bias; include data on which selection was based and genetic relationships among individuals

Steps in genetic evaluations

- Use of appropriate software to estimate breeding values and accuracy (e.g. Mix99, PEST, etc.)
  - Pertain to individuals for each trait
  - EBVs are BLUP
- Validating the estimated breeding values
  - Correlation between previous and current EBVs and regression of the current evaluation on the previous. Correlations are expected to be between 0.95 to 0.99 and the regression coefficient should be unity or very close, indicating there is no bias in our prediction
  - Cross-validation with independent datasets; split data into subsets (e.g. older vs. more recent)

Synopsising model progression

$Y = \alpha + \beta * X$ simple regression (statistics courses)

$y = Xb + e$ …… same in matrix/vector notation,

…… as discussed in this course

…… may now include multiple fixed effects

…… both, covariates and class variables

$y = Xb + Zu + e$

$y = Xb + Zu + Wc + e$

$y = Xb + Zu + Wm + Sc + e$

} mixed models including …… fixed and random effects

All above can be expanded to multivariate analyses (multiple traits, y)

By the end of Lecture 4

- Mixed model
  - Random vs. fixed effects
- Building a quantitative model
- Effect estimation
  - Genetic evaluation
- Variance de-composition
  - (Genetic) parameter estimation (e.g. trait heritability)
- Model utility

ANY FINAL QUESTIONS??